To Whom It May Concern,

I am happy to write this letter of recommendation for Ms. Misha Saraf. Misha is an outstanding student. I was her Mathematics teacher at Ramniwas Bajaj English High School, and I can attest to the fact that she is one of the few students who has strong skills and commitment to the subject. In class, I have always seen her being excited to tackle challenging topics like Algebra and Statistics. She also loves looking at problems from different and unexpected angles, and this, in turn, helps her classmates think more critically as well. Her assignments are always detailed, and I have noticed she has a clear understanding of the fundamentals.

In class tests as well as discussions, I have noticed that Misha took a very careful approach to the way she was studying the subject. She rarely left anything unchecked and often reviewed problems several times before finalizing her answers. She picks up new concepts easily, but what stands out is her eagerness to analyze them critically. In one of the tests, I posed a challenging question: A point robot moves in the plane from A = (0,0) to B = (1,0). There is a circular obstacle of radius r centered at (½, 0). The robot must move from point A to B with a path of the shortest possible arc-length that does not intersect the interior of the circle. Find the shortest part and its length for $0 < r < ½$. Most students begin by guessing or sketching paths, except for Misha! She first approached me, and we discussed why the shortest collision-free path must be made of straight segments joined by a circular arc, using symmetry about the x-axis. This immediately reduced the problem to analysing a single tangency point in the upper half-plane. She then chose coordinates for the tangency point T in terms of an angle theta, rather than fixing numbers early. What stood out most was the way she framed her concepts - step-by-step. She briefly connected the geometry to ideas from optimal paths and robot planning, even though the methods she was using

were still classical. I can seldom find students in class who can balance between traditional Euclidean reasoning and modern applications; this level is rare. This speaks strongly of Misha's mathematical ability and problem-solving depth.

Misha also has an impressive ability to visualize how complex problems can be broken down simply into manageable terms. Quite recently, I recall assigning a challenging integration problem to the class: A robot has a prior probability p that an obstacle is present. Two conditionally independent sensors report detections: Sensor A: P(detect | obs) = 0.90, P(detect |no obs) = 0.05. Sensor B: P(detect| obs) = 0.80, P(detect | no obs) = 0.10. If both sensors detect an obstacle, find the posterior probability that the obstacle is present. She did not jump directly to try substitution like other students, but rather she worked first to find out the structure of the problem: a prior belief, two noisy sensors, and the need to update the belief after receiving evidence. She started with the assumption of conditional independence, a step missed by most students, even though it was an integral part of multi-sensor fusion. This showed me that she understood why Bayes' theorem works, and was not just memorizing its applications. She then split the argument into two likelihood cases. When the obstacle was present, she multiplied the true positive rates and obtained P(A = 1, B = 1 | obs) = 0.72. When the obstacle was absent, she calculated the joint false positive probability as 0.005. She kept the reasoning structured and easy to follow. Throughout the solution, Misha linked textbook probability to real sensing tasks. She treated belief as something that updates only when new evidence arrives. This way of thinking matters for advanced work in probability, artificial intelligence, and robotics, and it shows strong grounding in fundamentals.

In summary, in my class, Misha has been an exceptional student who actively participated in all aspects of school life. I can confidently rank her among the

top 10% of students I have taught in my career. She has great academic potential and is a strong character admired by her peers. I believe her passion for Mathematics will lead her to a successful career in STEM, especially in software development and coding. It is with great pleasure and confidence that I recommend her.